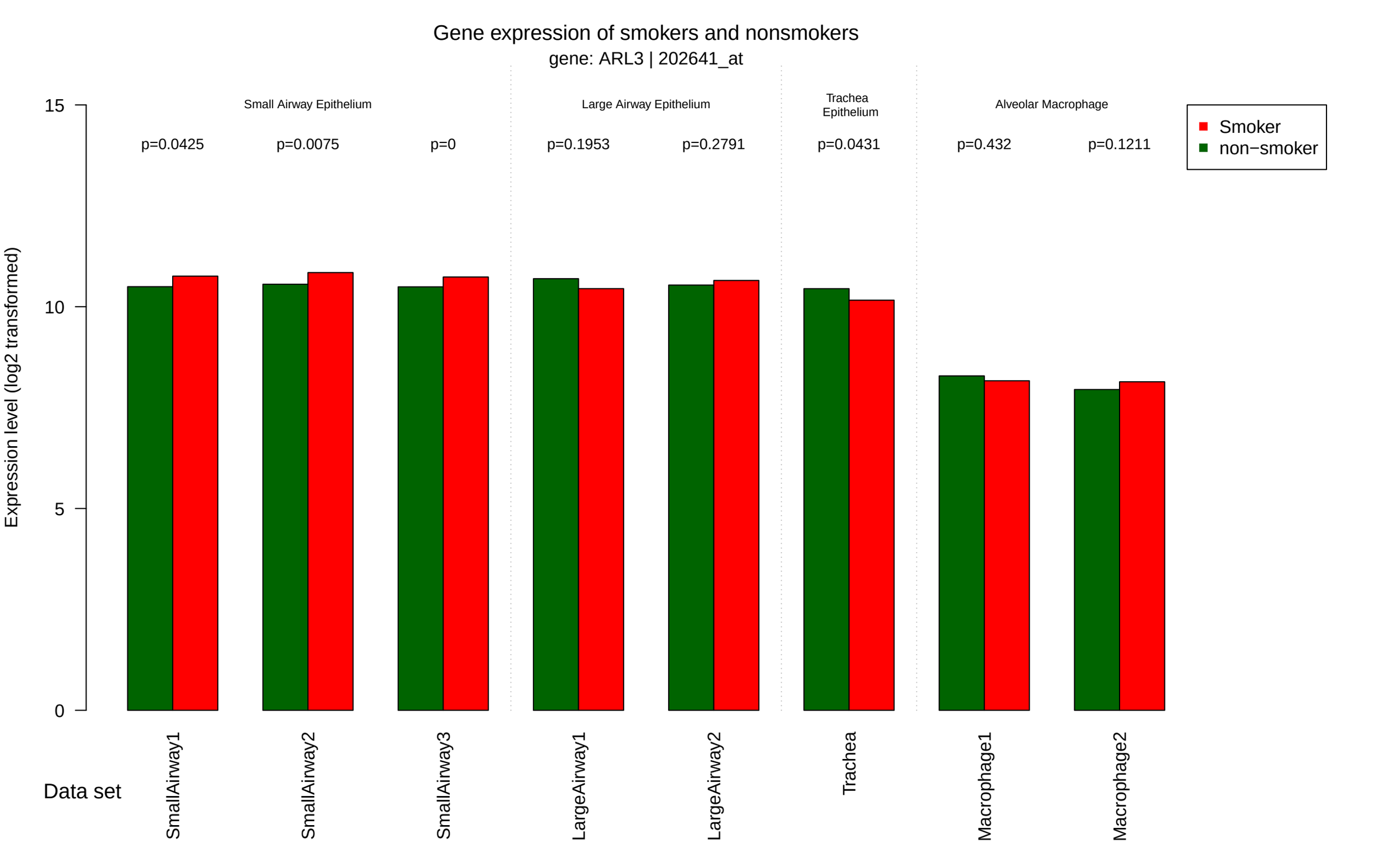
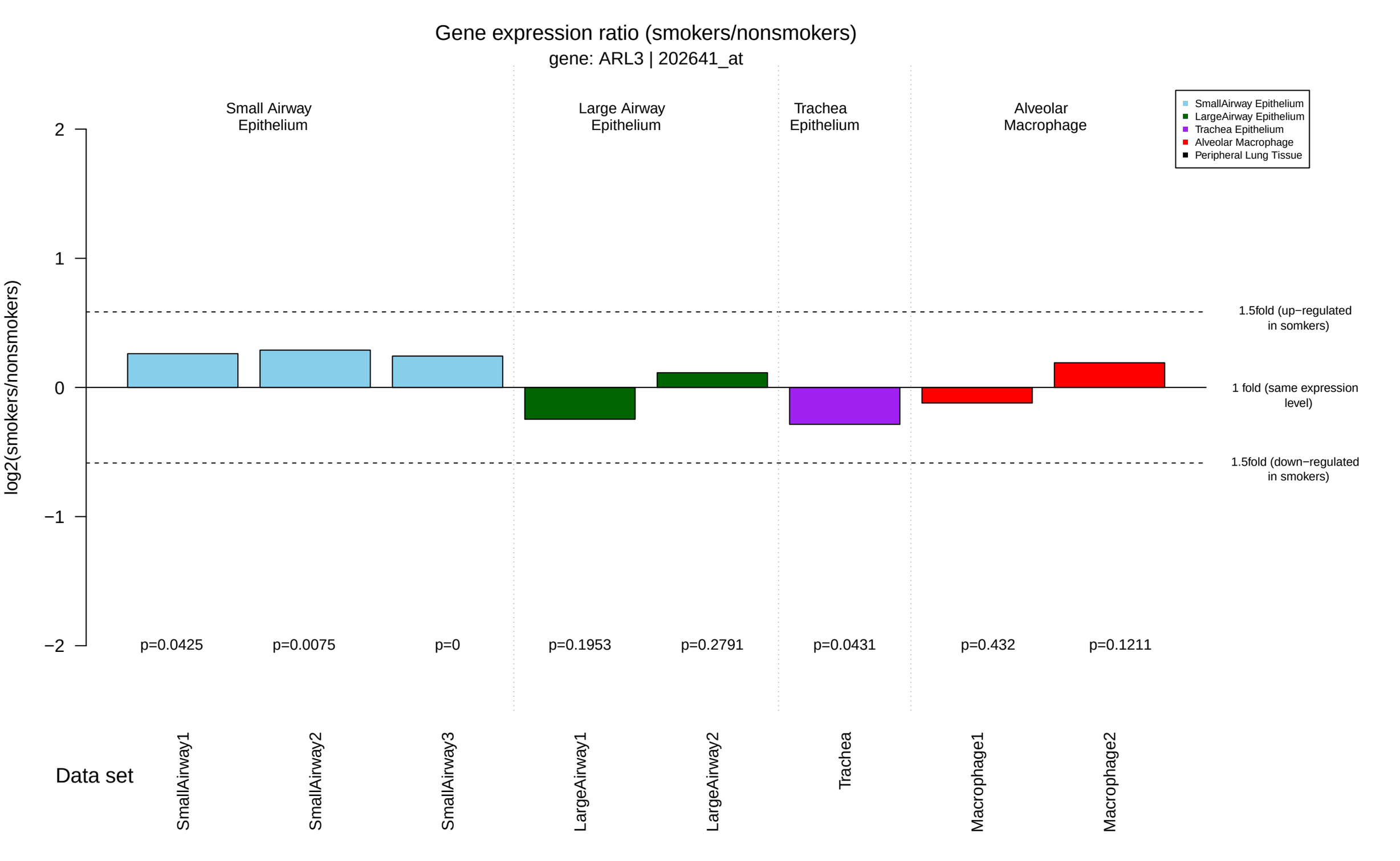
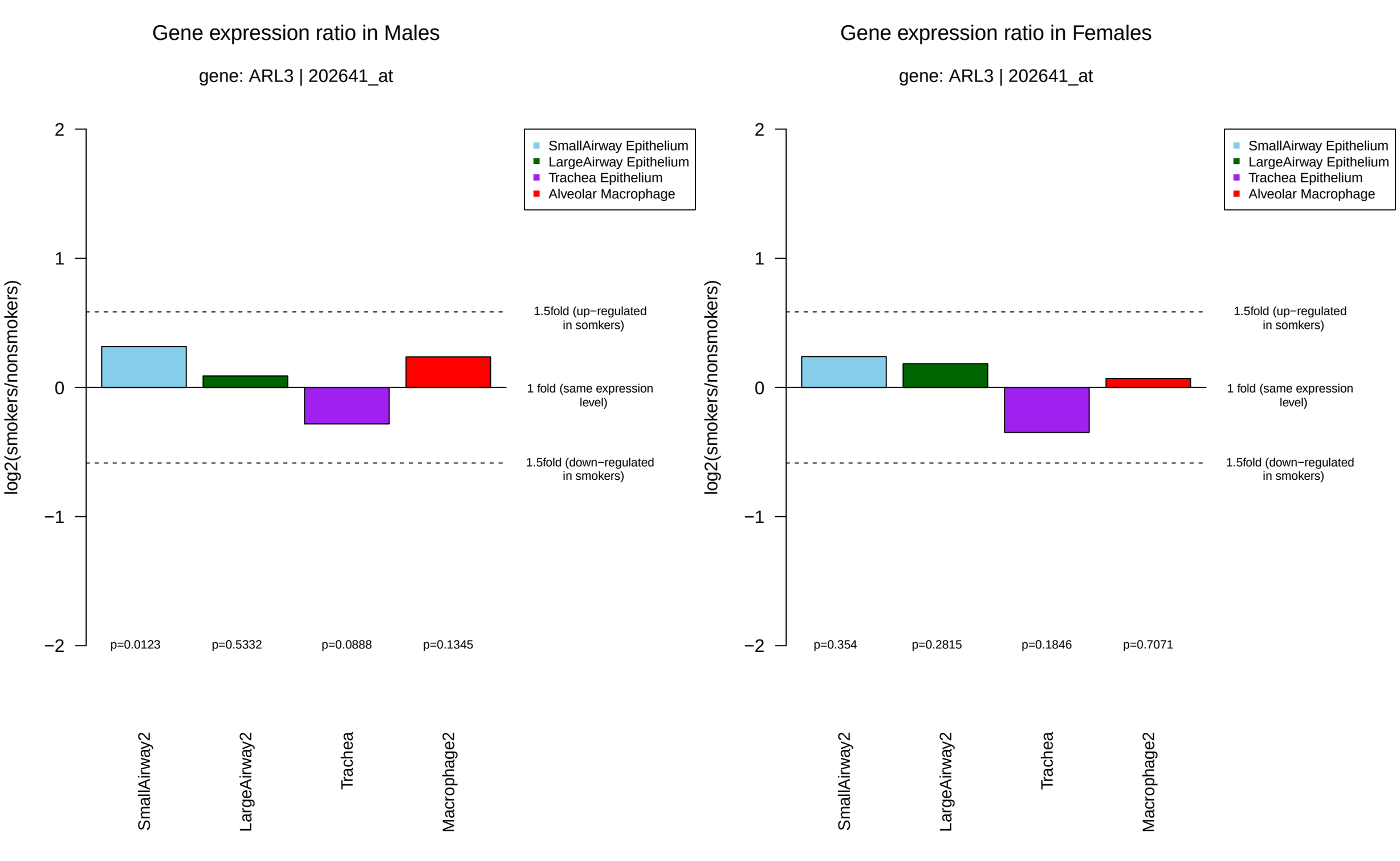
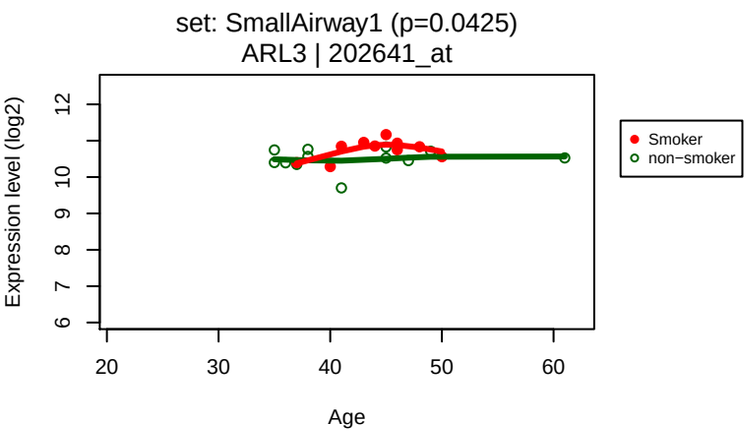
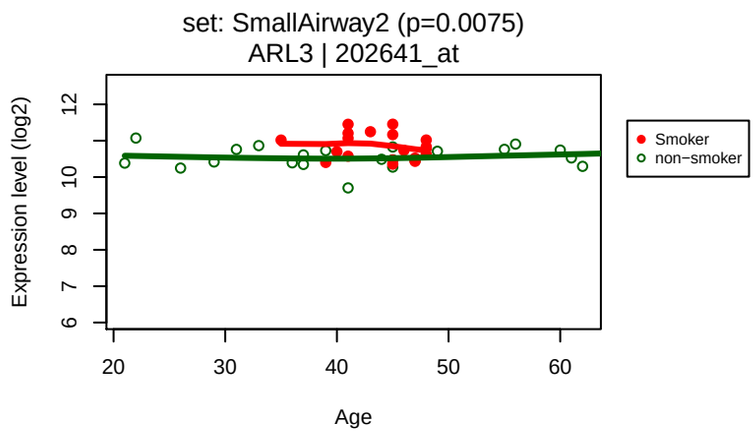
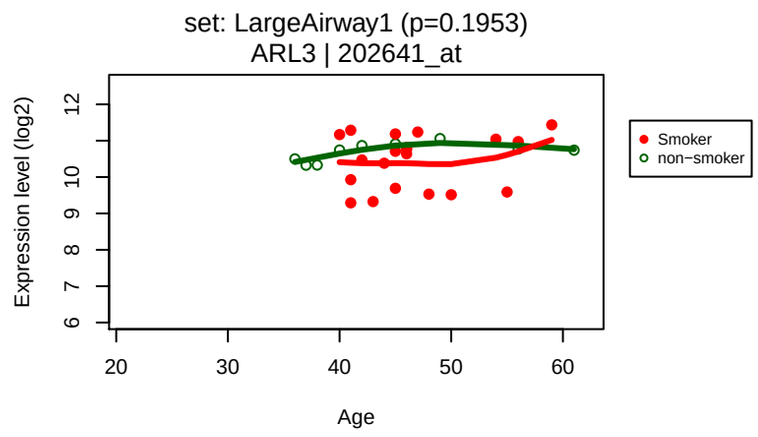

Gene expression of smokers and nonsmokers
gene: ARL3 | 202641_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0425
p=0.0075
p=0
p=0.1953
p=0.2791
p=0.0431
p=0.432
p=0.1211
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

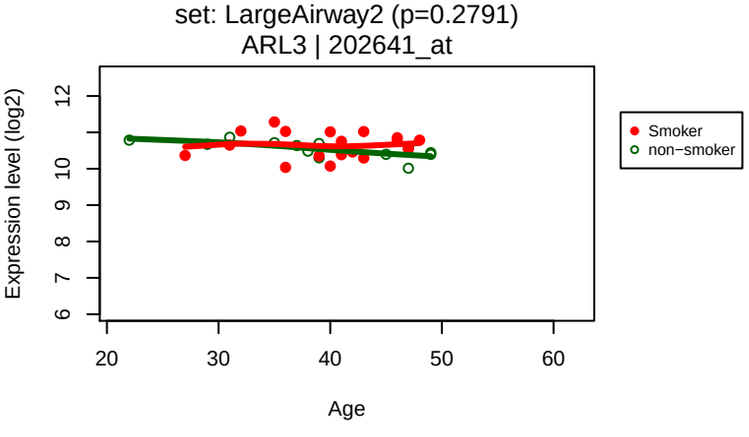

Gene expression ratio (smokers/nonsmokers)
gene: ARL3 | 202641_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0425
p=0.0075
p=0
p=0.1953
p=0.2791
p=0.0431
p=0.432
p=0.1211
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

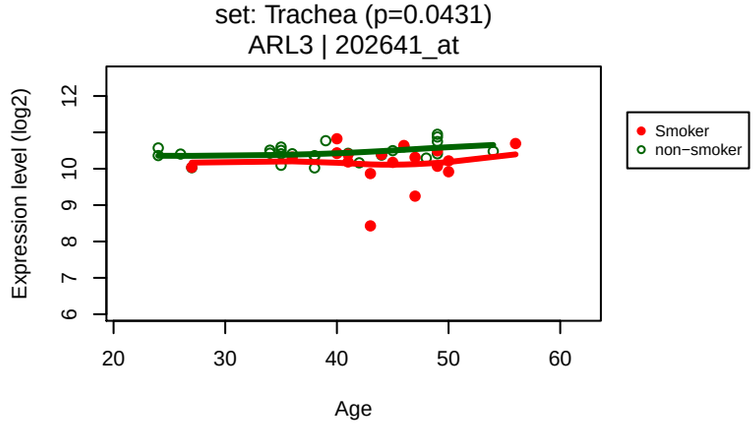

Gene expression ratio in Males
gene: ARL3 | 202641_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0123
p=0.5332
p=0.0888
p=0.1345
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL3 | 202641_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.354
p=0.2815
p=0.1846
p=0.7071
SmallAirway2
LargeAirway2
Trachea
Macrophage2

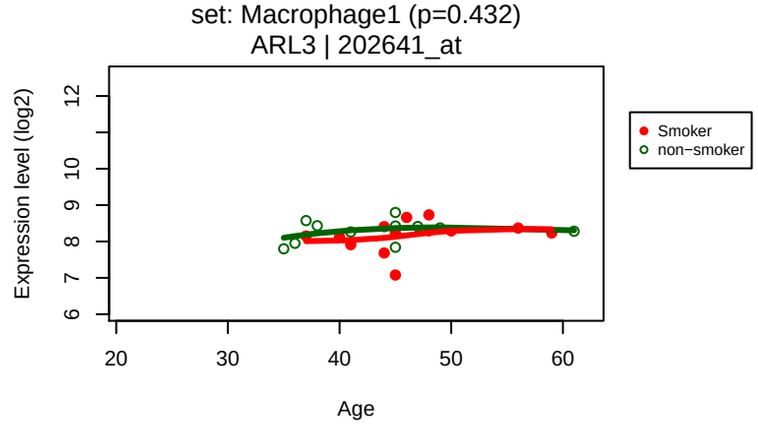

set: SmallAirway1 (p=0.0425)
ARL3 | 202641_at
Expression level (log2)
Age
Smoker
non−smoker

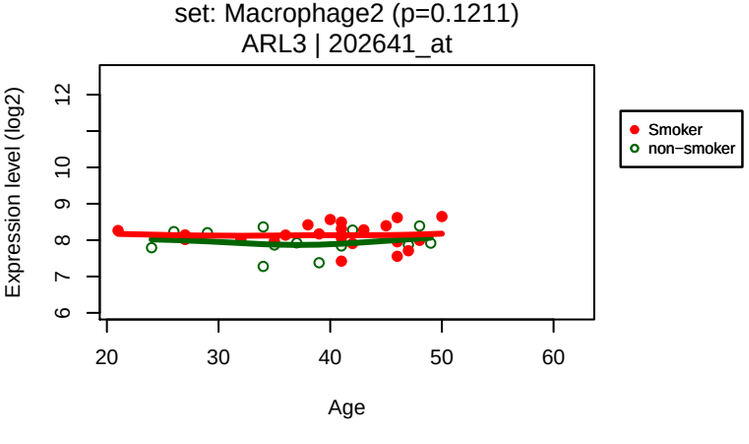

set: SmallAirway2 (p=0.0075)
ARL3 | 202641_at
Expression level (log2)
Age
Smoker
non−smoker

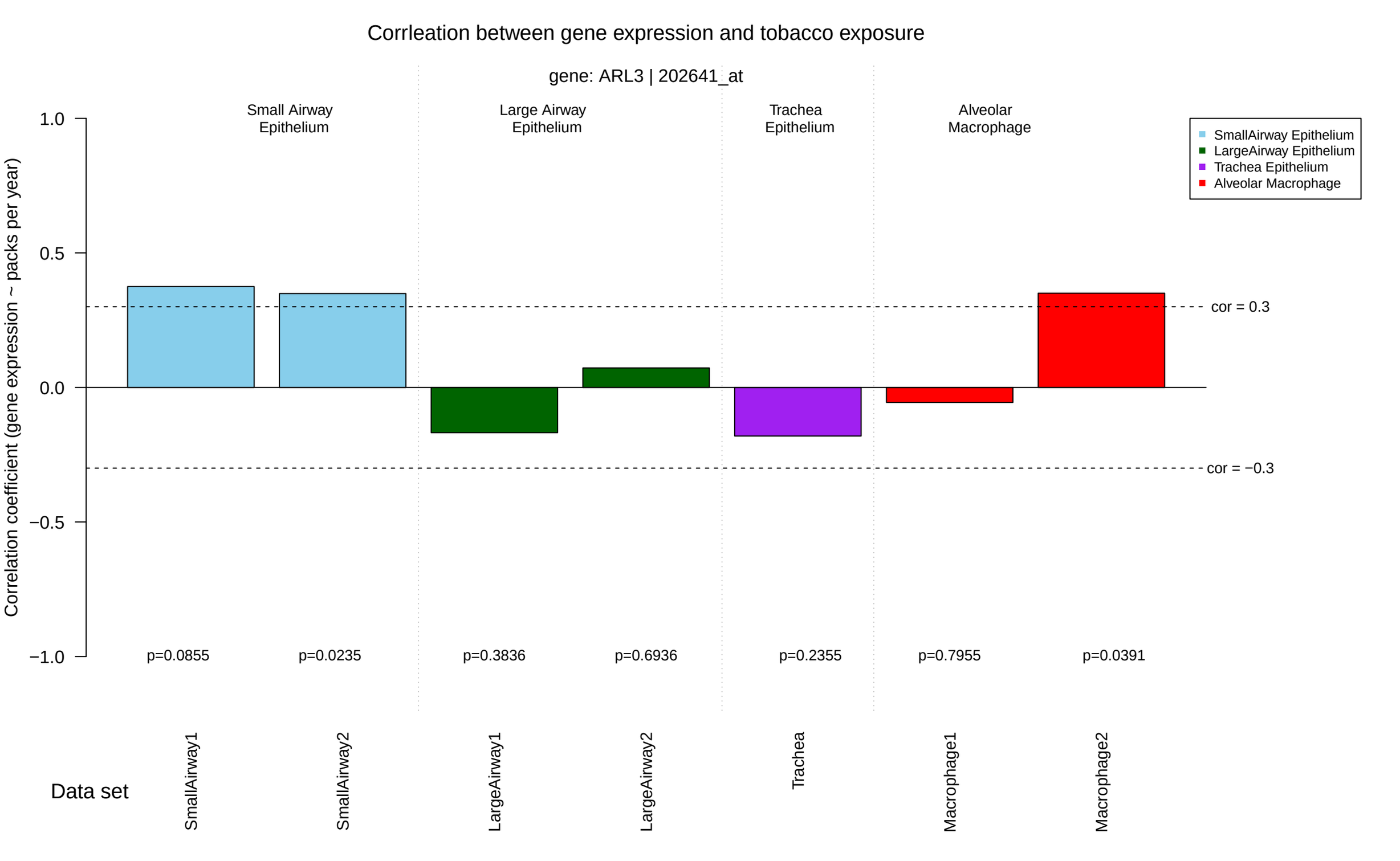

set: LargeAirway1 (p=0.1953)
ARL3 | 202641_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2791)
ARL3 | 202641_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0431)
ARL3 | 202641_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.432)
ARL3 | 202641_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1211)
ARL3 | 202641_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL3 | 202641_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0855
p=0.0235
p=0.3836
p=0.6936
p=0.2355
p=0.7955
p=0.0391
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2